

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab : Embedded Systems
Semester	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	01
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	Study of Embedded Microcontrollers, Development Environments, and Simulation Tools
Theory (100 words)	An embedded system is a specialized electronic system designed to perform a particular function efficiently. It consists of hardware components and software programmed to control the system. A microcontroller acts as the main control unit and includes a processor, memory, input/output ports, timers, and communication interfaces on a single chip. Examples include Arduino, 8051, AVR, PIC, ARM, and ESP32. A development environment provides the necessary tools for writing, compiling, debugging, and transferring programs to a microcontroller. Arduino IDE, Keil, MPLAB, and PlatformIO are commonly used environments. Simulation tools such as Proteus, Tinkercad, and Multisim allow circuits and programs to be tested virtually before implementing them on physical hardware. They help reduce development time, cost, and hardware-related errors.



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Procedure and Execution (100 Words)	1. Study the basic concepts and working principles of embedded systems and microcontrollers. 2. Identify commonly used microcontrollers such as 8051, Arduino, PIC, AVR, ARM, and ESP32. 3. Study the major features, specifications, and applications of the selected microcontroller. 4. Install and launch a suitable Integrated Development Environment (IDE) such as Arduino IDE or Keil. 5. Create a simple program, such as an LED blinking program, for the selected microcontroller. 6. Compile the program using the IDE and check that it is free from syntax and compilation errors. 7. Open a simulation platform such as Proteus or Tinkercad. 8. Design the required circuit by connecting the microcontroller with components such as an LED, resistor, power supply, and ground. 9. Load or run the compiled program in the simulated microcontroller. 10. Start the simulation and observe the behavior of the connected LED or other output device. 11. Compare the obtained output with the expected result and identify any errors in the program or circuit. 12. Correct the errors, if any, and run the simulation again to verify proper operation. 13. Record the circuit output, observations, and results in the practical record. 14. Conclude the practical by verifying the basic process of programming, simulating, and testing an embedded system.

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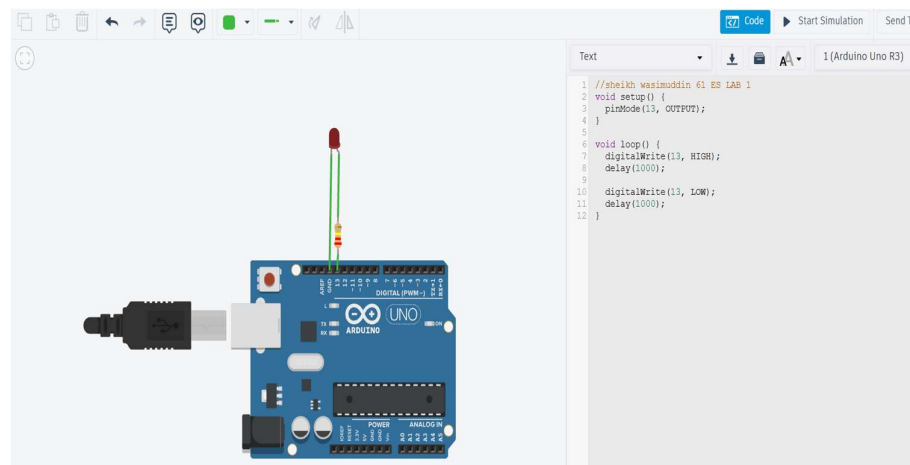
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Stepwise Screenshots with steps:

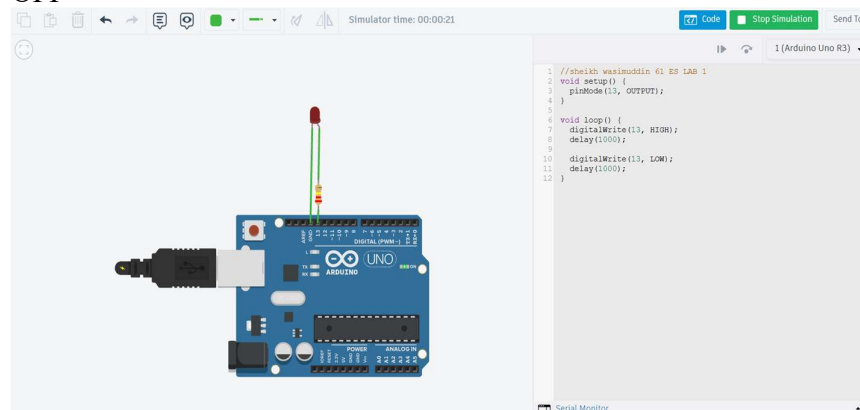
```
1 //sheikh wasimuddin 61 CSEIOT ES lab1
2 void setup() {
3   pinMode(13, OUTPUT);
4 }
5
6 void loop() {
7   digitalWrite(13, HIGH);
8   delay(1000);
9
10  digitalWrite(13, LOW);
11  delay(1000);
12 }
```

Output

Sketch uses 269688 bytes (20%) of program storage space. Maximum is 1310720 bytes.
Global variables use 22060 bytes (6%) of dynamic memory, leaving 305620 bytes for local variables. Maximum is 327680 bytes.



OFF





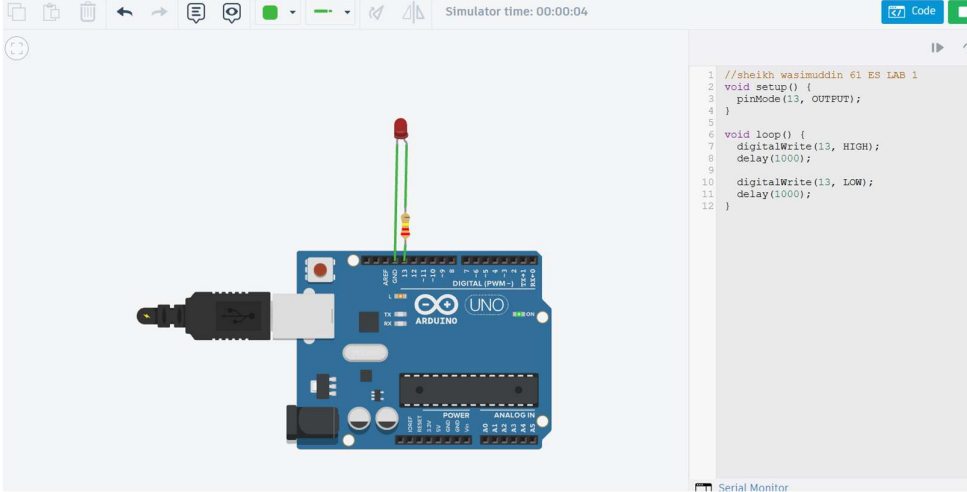
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	ON 
Conclusion	The practical successfully demonstrated the fundamentals of embedded microcontrollers, development environments, and simulation tools. An Arduino-based LED blinking application was developed using the Arduino IDE and successfully compiled without errors. The circuit was designed and tested using a simulation environment, where the LED was observed turning ON and OFF at regular intervals. This practical provided a basic understanding of microcontroller programming, circuit interfacing, program compilation, and simulation. It also helped in understanding how embedded systems can be developed and tested before implementation on physical hardware.
Date	20-08-2026